

It must be noted, though, that bioinformatics is still very niche in India, with only a handful of companies carrying out research that requires the use of bioinformatics. It is primarily large companies doing outsourced work from multi-national pharmaceutical companies that recruit. Also remember that bioinformatics is an application-oriented subject and demands a clear understanding of biological and allied concepts, and that students therefore need to build a research-oriented mindset.

Syllabi

Until recently, due to the specialised needs of the bioinformatics industry, there was no fixed course one could opt for. Companies employed graduates who had completed an M.Tech. and had a little background in biology, or those who had a B.Sc. background with a flair for programming. The bioinformatics courses available today are a mash-up of the B.Tech. and B.Sc. syllabi, covering concepts in programming and in biology. These days, to get into the industry, you need to have a B.Tech. or similar qualification. This is a stepping stone into the industry, to get further; you need a Master's degree with a super-specialisation in the research area you've opted for.

Due to media hype and a rush of students wanting to take up bioinformatics courses, there are many fly-by-night colleges, not affiliated to any reputed university, that offer courses in bioinformatics. Most equipment for biotech / bioinformatics courses is expensive; it is therefore important to check for lab facilities. It is also important to check the background of the faculty members of the college you are applying in before registering for a course. If you can, speak

Getting An Education	
Undergraduate Courses	URL
Amity Institute of Biotechnology—B.Tech.	http://tinyurl.com/2vxdkf
Vellore Institute Of Technology—B.Tech.	http://tinyurl.com/3a5k74
Tamil Nadu Agricultural University—B.Tech.	http://tinyurl.com/2o6lee
Sathyabama Institute of Science and Technology- .Tech.	http://tinyurl.com/3dhd4v
Postgraduate Courses	
Pune University—M.Sc. in Bioinformatics	http://tinyurl.com/33jc93
Jamia Millia Islamia University—M.Sc. in Bioinformatics	http://tinyurl.com/3ce52d
Tamil Nadu Agricultural University—M.Sc. in Bioinformatics	http://tinyurl.com/2lzbaj
Madras University—M.Sc. in Bioinformatics	http://tinyurl.com/2olfd6
University of Hyderabad—M.Tech. in Bioinformatics	http://tinyurl.com/36tq2f
University of Madras—M.Sc. in Bioinformatics	http://tinyurl.com/2kpqqe
Advanced Diploma Courses	
Pune University—Advanced Diploma in Bioinformatics	http://tinyurl.com/3aohj8
Madurai Kamaraj University—Advanced Post-graduate Diploma	http://tinyurl.com/2p5q3k
University of Hyderabad—Post-graduate Advanced Diploma in Bioinformatics	http://tinyurl.com/2oo6m3
Bioinformatics Institute of India—Post-graduate Diploma in Bioinformatics	http://tinyurl.com/2moz69
Ph.D. Research Courses	
IMTECH—Protein Science and Engineering	http://tinyurl.com/3dq5p9
CCMB—Bioinformatics	http://tinyurl.com/2ofeyp
NII—Bioinformatics	http://tinyurl.com/2jjv73
IGIB—Genome Informatics	http://tinyurl.com/3exfgy
CDFD—Computational Biology	http://tinyurl.com/3y2oc7
IICB—Structural Biology and Bioinformatics	http://tinyurl.com/2zloyb
IICT—Research Projects	http://tinyurl.com/3dndxp

Money Matters		
Position	Educational Pre-requisites	Remuneration (PA)
Junior Research Fellow	B.E., B.Tech., or B.Sc.	0.5 to 1 lakh
Senior Research Fellow	M.E., M.Tech., or M.Sc.	2 to 3 lakh
Quality and Data Analyst	M.E., M.Tech., or M.Sc.	2 to 3 lakh
Head Researcher	Ph.D. in your field of super-specialisation	4 to 7 lakh
Project Head	Ph.D. in your field of super-specialisation, in-depth knowledge of BioPerl and GMOD/Chado based applications	7 to 9 lakh

to alumni of the college regarding the lab facilities and such.

MBA degrees specialising in bioinformatics is a new phenomenon. Since these courses have only recently started, it is important to remember that their marketability has not been proven.

Head-starts

The computational software that most bioinformatics companies use runs on UNIX-like operating systems like Red Hat Linux and Fedora; it is therefore good to have hands-on experience with these while pursuing your graduation.

You can take up summer apprenticeship courses during the holidays or between semesters. These courses help you gain an understanding of how a bioinformatics project works, and can be an add-on to your résumé. They normally last for between three and six months. You could apply for a summer apprenticeship at certain institutions.

The Remuneration

Working in this industry requires patience and analytical skills. The ability to analyse statistical data from graphs and statistical tables is an added bonus.

According to Dr S K Mahajan, Gharda Professor of Biotechnology, Institute of Science, Mumbai, “The field of bioinformatics is demanding as well as low-paying. In spite of the number of students graduating from these courses every year, only a select few are offered lucrative placements.” According to Raghothama Chaerkady, Research Fellow at McKusick-Nathans Institute of Genetic Medicine, Johns Hopkins University, a competent bioinformatician assisting in research can make up to Rs 12,000 to 16,000 as a Junior Research Fellow. With a Master's degree, working on Proteomics, mass spectrometry based sequencing, microRNA, and such, it could be Rs 40,000 per month, and at the Ph.D. level, around Rs 40,000 to 80,000 per month.

When a company first recruits you, you start as a Junior Research Fellow. Here you would be in charge of assisting Senior Research Fellows in conducting tests on preset parameters of the project. This work is, for the most part, very basic and repetitive in nature. After three years of experience, you would be eligible for the posi-

Indicative Terms

Human genome: The genetic map of the human body's DNA, including chromosomes and genes. Mutation: A permanent structural alteration in DNA; in most cases, this refers to DNA changes done by researchers to improve an organism's chance of surviving. Modelling: Refers to the theoretical methods and computational processing used to mimic the behaviour of biological molecules. Proteomics: Roughly speaking, the study of proteins. Mass Spectrometry: A technique used	to determine the composition and abundance of the atoms in a molecular substance. Sequencing: A technique for identifying nucleotide sequences in a DNA fragment. MicroRNA: A short piece of single-stranded RNA that controls the expression of genes. Docking: Computational exploration of the possible binding modes of DNA Biostatistics: The application of statistics to a wide range of topics in biology, medicine and to agriculture in particular.
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Though we don't actively look for students from “bioinformatics courses,” candidates who have a B.Tech. or M.Sc. in any branch of the life sciences are preferred”

Dr M Vidyasagar
Executive vice president (Advanced Technology)
Tata Consultancy Services

tion of Senior Research Fellow. In this role, you would be in charge of suggesting the tests and research paths to follow in order to reach the goals of the project. For example: a project related to the design and analysis of DNA microarrays, with the goal being to extend an existing Java application that models, say, the behaviour of synthetic DNA under certain circumstances.

Most biotechnology companies in India today work on techniques involving docking, modelling, sequence analysis, and advanced biostatistics. You can also branch out to being a Quality and Data Analyst, in charge of overseeing the data generated by the various tests and validating their authenticity.

After another three to four years of experience, you could qualify to be a Head Researcher—in charge of providing inputs to the project in your field of specialisation. After two years, you could aim to become Project Head. This role would require you to have in-depth knowledge in your field of specialisation, and thorough experience on the software tools used in the bioinformatics industry; you would be expected to lead—and define a roadmap for—a project that falls within the scope of sequence assembly, database design and maintenance, sequence analysis, and Proteomics.

The table overleaf lists the educational background required for each position in the industry, with an estimate of the remuneration one could expect.

Parting Notes

The bioinformatics industry will continue to evolve into a highly specialised sector that will service biotechnology companies. Solutions developed will help researchers sort and retrieve data more effectively. For the industry to grow in India, global pharmaceutical companies need to conduct their research and development activities here, rather than just doing field trials, which is mostly the case today.

The science is constantly evolving; students need to constantly update themselves on the latest trends, as well as the software and techniques used. ■

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